Physics Activities

Fast-Paced Competitions in the US

- The $F = ma$/USAPhO exams are the premier high school physics competitions in the US. The $F = ma$ exam is a fast-paced multiple choice exam (75 minutes, 25 questions) focusing on tricky mechanics problems, while the USAPhO exam has longer problems which require written solutions (3 hours, 6 questions).
- The PhysicsBowl is a fast-paced multiple choice exam (45 minutes, 40 questions) at the level of AP Physics 1 and 2, mixed with trivia and history questions.
- The Science Bowl is an exciting buzzer-based team competition covering all fields of science. To do well as a team's physics player, you need to be able to solve AP Physics 1 and 2 questions *extremely* quickly, and also have a good deal of general physics knowledge. Or, if you have a lot of general knowledge about science, consider being the science player on a quiz bowl team.
- Your state may have local science competitions. For example, in New Jersey you can participate in the New Jersey Science League.
- The USAAAO (Astronomy Olympiad) covers a variety of fascinating topics. You'll need dedicated astronomy study to be competitive, but physics knowledge provides a good foundation.

Longer-Term Competitions

- The Online Physics Olympiad is a team-based, multi-day competition written largely by high schoolers, founded in 2020. It is one of the largest international physics competitions, with hundreds of teams participating. Because it's student written, the problems can be contrived or ambiguous, but they're also rapidly improving over time.
- The Rudolf Ortvay competition is a marathon competition with several dozen questions to be solved in 10 days, requiring undergraduate physics knowledge.
- The High School Mathematical Contest in Modeling and MathWorks Math Modeling Challenge are team-based competitions (with time limits of 14 days and 14 hours, respectively) where students develop mathematical models for an open-ended real-world problem.
- KoMaL and FYKOS are Hungarian and Czech contests for high school students which issue a problem set every month. Their problems tend to be elegant and instructive.
- The Physics Cup is an extremely difficult competition with a monthly question and weekly hints. The problems are clearly posed and very instructive, requiring the student to come up with a great deal on their own. In theory, only introductory physics is required, but exposure to more advanced math, and background in Euclidean geometry tends to help.
- The USIYPT is a team-based, year-long competition where students investigate open-ended *experimental* physics problems, and defend their results in two days of scientific debate. Out of all the competitions listed here, this one gives students the most freedom by far, but it requires more to get started. It helps if your high school has an existing physics club, and lab equipment.
- CERN's Beamline for Schools program gives teams of high school students an opportunity to perform an experiment with a real particle accelerator.

Summer Programs

- PRIMES is an excellent program for pure math research, which matches high school students with good mentors for free. Admissions are relatively transparent, requiring the solution of a problem set; Olympiad qualifications help. Also see the Euler Circle's advice.
- RSI is the best program for general science research, and it's also free. However, it's extremely selective; you generally need to *already* have research experience to stand a chance of admission, especially if you're from a large state.
- SPARC is a free program which covers "applied rationality", which means an array of fun topics such as game theory, economics, and cognitive science. Though it doesn't have much to do with physics, it attracts many Olympiad students and is a lot of fun, with plenty of free time for discussions. A major part of the application is a set of interesting, open-ended questions.
- SSP Astrophysics and YSPA are fun programs for high school juniors; students take real data in observatories overnight. Both are expensive, but offer substantial financial aid.
- QSYS is a free online program run by the University of Waterloo, introducing quantum information; Canadians can visit in-person for a laboratory day.
- QuarkNet is a set of local programs distributed across the United States, which helps high school students get involved with particle physics; check its map for programs in your area.
- If you're part of an underrepresented minority group, there are several excellent, free programs which will let you do research alongside similar students, such as SAMS, MITES, and YRP.
- Avoid unknown programs that promise a college admissions payoff at a high price. For example, in 2020, the Summer STEM Institute charged high school students almost $10,000 for a few hours of Zoom calls with college undergraduates, claiming they would reveal the secret to award-winning research. Other programs run their own journals, which will accept just about anything. These programs are a trap for unwise parents. They embody everything wrong with the American science fair and college admissions process.

Starting Research

Most science fair winners get their projects by reaching out to professors.

- The hardest stumbling block is realizing this is a sensible thing to try. It's unlikely that you could do *anything* more efficiently than a professor or even a graduate student, since you're missing years of knowledge. But professors really do choose to mentor high school students sometimes, for several reasons: you have more free time, you could become useful, they could have a postdoc or grad student who wants a helper, and they might like you for some reason.
- If you're lucky enough to already know professors, asking them in person is far better than anything else. The next best option is to be lucky enough go to one of the few top high schools which have established programs to match students with mentors.
- Otherwise, to get started, browse the faculty listings at nearby universities. Read their biographies and websites and see what sparks your interest. Focus on friendly faculty who care about mentoring; a professor happy to work with undergraduates, and who publicly encourages them to apply, is much more likely to be willing to work with high school students.

- Don't be overly picky. For example, don't fall into the trap of insisting on a string theory project because of its glamour. Also don't laser focus on the highest ranked university within a thousand miles. Cutting-edge research is done at hundreds of universities in the United States.

- Pick several professors at a time to cold email. Don't be demanding; be aware that you're asking for a big favor. Your first email should briefly convey the following information:

  - Your interest in physics. Don't try to exaggerate here, because it will be obvious if you do. For example, if you've learned some quantum mechanics by reading a textbook, that's a great thing to mention. But don't say you've been dying to work on superconducting spintronics, or whatever it is the professor works on, since infancy. A decent number of professors have heard about Olympiads, so mentioning your awards can help.
  - Your ability to code, or more generally to make things. For a high school student to be helpful at all, the professor needs to be able to find a specific, reasonably sized task for them to do, requiring less technical background. In most cases this means you'll end up coding. Highlight any impressive projects or artifacts you've made.
  - The amount of time you can commit. Summer tends to be a good time to start.

  No matter what your qualifications are, the vast majority of requests will be ignored; that's normal. Remember, you only need one!

- Professors typically skim hundreds of emails a day, so you should minimize the energy it takes for them to parse your email. For example, don't send a multi-page resume, or a link to a fancy vibecoded website with no content on it. You can ask AI for advice, but never direct AI to expand a short request into a long email; it will simply insert the same generic fluff the professors have seen a thousand times before. They will immediately notice, get annoyed at how you're taking up their time, and delete your email.

- These days, some less scrupulous students are sending hundreds of emails to professors at once. To avoid getting lumped in with them, think hard about why you're emailing every person you are, and have a reason in mind why they should reply to you, rather than all the others. For example, perhaps their university is close enough for you to regularly show up, or you read and deeply understood some of their popular science work.

- Here are some additional points of email etiquette:

  - In the United States, titles generally don't matter as much as in Europe or Asia, but it's still best to address professors as Prof. Lastname and postdocs as Dr. Lastname in a first email. If they reply to you and sign off with their first name, feel free to use it.
  - You don't need a long email signature; just sign off with "Thanks", followed by your name.
  - If you don't get a response in a week or two, your email could have just fallen down in their inbox, so you can send a reminder. Don't send multiple reminders, and don't demand a response; if they're too busy to respond, they're also certainly too busy to mentor you.

- Remember that research takes a long time, and don't obsess about jumping to a final result for the science fair. You'll probably need weeks of reading and listening just to figure out the basics of what your problem is about. Sometimes, months of work will yield nothing at all!

If you stick around in your research group, you'll find that the real benefit of mentorship has nothing to do with winning the science fair. You'll gain a much deeper appreciation of how science works by being part of a real scientific community.

A Rant About Science Fairs

High school science fairs are a long-standing American tradition. Winners of the leading ones, such as Regeneron ISEF/STS, receive a great deal of media attention, large cash prizes, and effectively guaranteed admission to top colleges, all while benefiting humanity with their results. But while I support giving high school students chances to do research, I feel the science fair system has growing flaws, which render it unhealthy for young students on the whole.

The first problem is that the judging is superficial. At the county and state level, judges rarely have specialized knowledge, and at the national level, they don't have time to thoroughly investigate each project. This is exemplified by the fact that recently, there have been winners whose results were completely falsified. In such a system, kids are incentivized to practice looking superficially impressive to adults, rather than developing their foundations and knowing what they don't know.[1] These problems are exacerbated by AI, which lets students produce projects they don't understand.

The second problem is that the system is inequitable. It's essential to find a professor to mentor you, but simultaneously very difficult. In practice, many students find such mentors through their parents' networks (and there are even cases where parents themselves do the projects), or from the resources of their well-established private school. There are admirable efforts to give research mentors to students outside these systems, but ultimately, they are still distributing a few lifelines at a time. They can't expand access for everyone, because the supply of mentors is very limited, and so success will be largely based on resources or luck. By contrast, in Olympiads you can succeed (as I did) by reading a $20 used textbook and thinking hard on your own; that is true equity.

The third problem is that it's taken too seriously. Many parents view it as a hackable path to elite college admissions, leading to bad behavior. For example, during my PhD at Stanford, my friends and I received constant offers to add high school students to our research projects, in exchange for thousands of dollars. It affects the students too. In the 2010s, science fairs were held at Google and the White House, and their winners were given glowing media articles and slickly produced videos, declaring them the future of science. So much praise isn't healthy for young physics students, who often haven't even learned quantum mechanics yet.

The fourth problem is that it gives a systematically wrong picture of how science is done. High school students need substantial guidance to do good research; even the best child prodigies are comparable to a new graduate student. But the press often leans into the same cliche, downplaying the role of the mentor and suggesting that the student came up with the whole thing, in a single flash of creative inspiration. It's what Cal Newport calls the failed simulation effect. The achievement looks more impressive the less clear it is how it arose, so everybody involved is incentivized to erase the intermediate steps. (The same is true, of course, for the occasional articles about preteens building "nuclear reactors" in their garage.)

When I was young, these "whiz kid" articles discouraged me. After all, I'd read several physics textbooks, yet I'd never had a flash of inspiration that turned into a breakthrough. I worried I was missing a creative spark, while in reality I was missing what was scrubbed out of the press release. There are many good paths to doing science, and if you learn physics well, the door will be open.

[1] I am also told by some past winners that at the top level, judges like to test knowledge by asking physics brainteasers. This is a nice touch, but it's far more ad hoc and less scalable than an ordinary written exam.